# A Representation-Centred Evaluation of Latent Diffusion for Afrikaans Text Generation

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**Abstract**

Diffusion models are a major approach to generative modelling, but adapting them to text is difficult because language is represented through discrete tokens while standard diffusion methods operate most naturally in continuous spaces. Latent diffusion addresses this mismatch by encoding text into a continuous representation, denoising in latent space, and decoding generated latents back into text. Its usefulness for language generation therefore depends on representation quality, tokenisation behaviour, corpus conditions, and evaluation design.

This proposed research will evaluate latent diffusion for unconditional Afrikaans text generation using the NCHLT Afrikaans Text Corpora. The core hypothesis is that, after English implementation validation, the adapted pipeline will learn a representation sufficient for reconstruction-quality generation and achieve mGPT perplexity no more than 20% higher than a matched autoregressive baseline. An English OpenWebText validation stage will first be used only as a supporting implementation sanity check before the Afrikaans adaptation.

The evaluation will use reconstruction loss for latent representation quality, mGPT perplexity for a limited external generation-quality signal, Distinct-1/Distinct-2 and repeated n-gram rates for degeneracy, and tokenisation fertility for representation and compute pressure. The intended contribution is a bounded diagnostic analysis for one concrete Afrikaans setting, packaged in a reproducible framework, not a claim that latent diffusion solves African-language generation generally.

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# Chapter 1

# Introduction

The cited diffusion-language literature motivating this proposal establishes that latent and continuous diffusion methods have been developed for language generation [Li et al., 2022; Gong et al., 2023; Lovelace et al., 2023], but it does not establish how such methods behave in African-language corpus settings. Moving from diffusion-based generative modelling to language generation is not a direct transfer of method; it is a problem of representation. Diffusion models have shown especially strong results in image generation, where data can be gradually corrupted with Gaussian noise and recovered through learned denoising over continuous pixel or latent representations [Ho et al., 2020; Rombach et al., 2022]. Text, however, is represented as sequences of discrete tokens, and applying diffusion to language requires deciding what representation should be noised and denoised before any generation can take place [Li et al., 2022; Lovelace et al., 2023]. This representational challenge is especially important for many African languages, where data scarcity, tokenisation inefficiency, uneven benchmark coverage, and limited infrastructure constrain what can be reliably learned from a corpus [Joshi et al., 2020; Nekoto et al., 2020; Lundin et al., 2025]. The question addressed in this proposal is therefore not only whether latent diffusion can generate text, but whether it can be evaluated as a credible method under the corpus, tokenisation, and evaluation constraints of a selected African-language setting.

Latent diffusion offers one principled response to the representational problem. Following the latent-diffusion principle developed for images [Rombach et al., 2022], language latent diffusion compresses text into a continuous representation using an encoder, performs diffusion in that compact latent space, and then decodes generated latent representations back into text [Lovelace et al., 2023]. This creates a useful division of labour: the autoencoder handles compression and reconstruction, while the diffusion model learns the distribution of latent representations. For language, however, this division is only useful if the learned representation preserves enough linguistic information. Sentence-level autoencoders and variational autoencoders can fail through posterior collapse or KL vanishing, where the decoder ignores the latent variable and the latent space becomes uninformative [Bowman et al., 2016; Li et al., 2020]. The proposed research therefore treats representation quality as a central object of evaluation rather than as a preliminary implementation detail.

The precise research gap is that the cited diffusion-language studies evaluate primarily in English or other high-resource benchmark settings [Li et al., 2022; Gong et al., 2023;

Lovelace et al., 2023]. Existing work shows that diffusion can be adapted to language through discrete diffusion, embedding diffusion, sequence-to-sequence diffusion, or latent diffusion, but it does not establish whether a latent-diffusion pipeline remains credible when adapted from an English working implementation to an Afrikaans corpus under honours-level compute and evaluation constraints. The transferability issue is not only language identity: corpus scale, tokenisation fertility, and the reliability of external evaluation models may change the reconstruction burden and the interpretation of generation metrics. This proposal addresses that gap through a bounded Afrikaans case study rather than a broad claim about African languages.

Following discussion with Dr Devon, the proposed research will begin with an English validation stage before moving to Afrikaans. This stage has a practical methodological role. It will confirm that the latent-diffusion implementation can train, generate, and be evaluated in the language setting for which the public implementation and much of the literature were designed. This reduces the risk of confusing implementation failure with Afrikaans-specific corpus, tokenisation, or representation effects. Only after the English pipeline is operational will the proposed research adapt the pipeline to the NCHLT Afrikaans Text Corpora. The English stage is therefore a methodological precondition and implementation control, not a contribution. The Afrikaans stage remains the focus of the hypothesis, research questions, and contribution.

Afrikaans is a meaningful first case because it has a documented NCHLT corpus, which makes a controlled honours-scale experiment feasible while remaining within the African-language NLP landscape [Nekoto et al., 2020; Eiselen and Puttkammer, 2014]. Resource coverage, evaluation infrastructure, and tokenisation behaviour still cannot be assumed to match high-resource English settings [Joshi et al., 2020; Nekoto et al., 2020; Lundin et al., 2025]. The proposed research will use the NCHLT Afrikaans Text Corpora as the primary Afrikaans data source [Eiselen and Puttkammer, 2014]. The selected NCHLT text component will be used for unconditional monolingual text generation. This task is deliberately narrow: the model will generate Afrikaans text without a conditioning input. Unconditional generation is the chosen starting task because it isolates representation quality without requiring parallel or reference outputs, which would introduce different data, baselines, and evaluation protocols. The study will record corpus statistics before training and will interpret all findings as specific to the selected corpus and task.

Throughout this proposal, $x$ denotes an input text sequence or its vector representation, depending on context. The clean input before noising is denoted $x_0$, and the noised version at diffusion timestep $t$ is denoted $x_t$. For a latent diffusion model, $E$ denotes an encoder, $D$ denotes a decoder, and $z = E(x)$ denotes the latent representation of an input sequence. The noised latent representation is denoted $z_t$, and the denoising model is denoted $\epsilon_\theta$, where $\theta$ represents trainable parameters. This unified notation keeps the descriptions of the English validation stage, Afrikaans adaptation stage, autoencoder, latent space, diffusion process, and evaluation consistent across the proposal.

The proposed research will make three intended contributions. First, it will provide a bounded empirical evaluation of latent diffusion for unconditional Afrikaans text generation using the NCHLT Afrikaans Text Corpora, after first validating the implementation on English text. Second, it will analyse how representation quality, SentencePiece BPE tokenisation fertility, corpus conditions, and automatic evaluation limitations affect latent-diffusion performance when moving from an English working pipeline to Afrikaans.

Third, it will provide an analysis of how these factors interact in one reproducible diagnostic framework, so that later work can identify whether failure comes from representation learning, diffusion sampling, tokenisation, corpus conditions, or automatic evaluation. The rest of this proposal is structured as follows. Chapter 2 gives the background needed to understand diffusion models, latent diffusion, and African-language generation constraints. Chapter 3 states the aim, objectives, hypothesis, research questions, data, model, experiments, baselines, metrics, and decision rules. Chapter 4 presents the timeline, deliverables, risks, mitigation strategies, and limitations. Chapter 5 summarises the proposed research and explains why the hypothesis matters for machine learning, natural language processing, and African-language technology.

# Chapter 2

# Background

This chapter provides the technical and contextual background required for the proposed research. It is organised around three questions. Section 2.1 explains what diffusion models learn. Section 2.2 explains why latent diffusion depends on representation quality. Section 2.3 explains how diffusion has been adapted to language and why African-language corpus, tokenisation, and evaluation constraints matter for the proposed Afrikaans case study. The chapter is not intended to reproduce a full literature review; it covers only the background needed to justify the proposed methodology.

## 2.1 Diffusion Models and Learned Denoising

Diffusion models frame generation as the reversal of a gradual corruption process. In denoising diffusion probabilistic models, a clean data sample $x_0$ is progressively corrupted through a forward process until it approximates pure noise, and a learned reverse process is trained to reconstruct less noisy samples step by step [Ho et al., 2020]. The forward process is defined as a Markov chain:

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}), \tag{2.1}$$

where $T$ is the final diffusion timestep. In the Gaussian formulation of Ho et al. [2020], the transition distribution adds a controlled amount of noise at each timestep:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I), \tag{2.2}$$

where $\beta_t$ controls the noise level at timestep $t$ and $I$ is the identity matrix.

The forward process can also be written in closed form by defining $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, which allows a noisy sample $x_t$ to be generated directly from $x_0$:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{2.3}$$

A neural network $\epsilon_\theta$ is then trained to predict the added noise $\epsilon$. The simplified denoising objective is:

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t,x_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]. \tag{2.4}$$

This framework demonstrated strong image-generation performance [Ho et al., 2020]. Later work reported improved likelihoods and more efficient sampling, including far fewer sampling passes when learning reverse-process variances [Nichol and Dhariwal, 2021]. These developments matter for the proposed research because they establish diffusion as a general generative framework, but they do not by themselves solve the problem of applying diffusion to discrete text.

## 2.2 Latent Diffusion and Representation Spaces

Latent diffusion changes where the diffusion process is applied. Instead of operating directly on raw observations, an encoder maps data into a lower-dimensional latent space, diffusion operates in that space, and a decoder maps generated latent representations back into the original domain [Rombach et al., 2022; Lovelace et al., 2023]. An encoder $E$ and decoder $D$ define

$$z = E(x), \qquad \hat{x} = D(z). \tag{2.5}$$

The diffusion model is trained on noised latent representations $z_t$ rather than on the original input $x$.

For images, latent diffusion reduces computational cost and allows denoising to operate in a compact representation space [Rombach et al., 2022]. For language, the same idea is attractive but riskier. A generated text sequence is only as good as the representation that the autoencoder learns and the decoder can reconstruct. If the latent representation does not preserve important linguistic information, then diffusion in latent space cannot recover it. This is why representation quality is not a secondary detail in this proposal; it is the first empirical question.

Text autoencoders also face the risk of posterior collapse. Bowman et al. [2016] identify the problem in sentence variational autoencoders: a strong decoder can learn to ignore the latent variable, leaving the latent space uninformative. Li et al. [2020] address KL vanishing by combining pretrained encoder and decoder language models and pretraining a meaningful latent space on large text corpora. Lovelace et al. [2023] avoid the standard variational formulation by freezing pretrained encoder-decoder language-model components, learning compression and reconstruction modules, and training diffusion over the resulting compact latents. These design choices motivate the proposed research's focus on reconstruction loss and sample inspection before interpreting diffusion results.

## 2.3 Diffusion for Language Generation in African-Language Settings

Applying diffusion to language requires a decision about what is corrupted and reconstructed. Existing approaches differ mainly in whether they operate over token-like structures, continuous embeddings, or learned latent representations. This proposal focuses on the latter two because they are closest to the implementation path used in the proposed work [Li et al., 2022; Lovelace et al., 2023].

One approach is to move text into continuous embedding space. Li et al. [2022] introduce Diffusion-LM, which applies diffusion to continuous word embeddings and then maps generated vectors back to tokens. For sequence-to-sequence generation, Gong et al. [2023]

introduce DiffuSeq, which applies noise to target-side embeddings. These methods show that diffusion can be adapted to language, but they remain sensitive to embedding quality, token recovery, and task-specific evaluation.

The third direction is latent diffusion for language. Lovelace et al. [2023] use a language autoencoder to map text into compact continuous representations, perform diffusion in that latent space, and decode generated latents back into text. This is the closest published methodological starting point for the proposed research. Because this work and much of the surrounding literature are evaluated primarily in English or high-resource settings, the proposed research will first validate the pipeline in English before adapting it to Afrikaans. The English validation stage is not intended to answer the African-language research question; it is intended to establish that the implementation is functioning before failures are interpreted in the Afrikaans setting. The specific transferability gap is that English validation may rely on larger corpus scale, lower tokenisation pressure, and better-calibrated evaluation tools, while the Afrikaans setting may change effective sequence length, latent reconstruction difficulty, and external perplexity reliability.

African-language NLP is shaped by structural constraints rather than only by smaller datasets. Joshi et al. [2020] show that NLP resources and technologies are unevenly distributed across the world's languages. Nekoto et al. [2020] extend this point in African machine translation, arguing that low-resourcedness includes data availability, curation conditions, infrastructure access, participation in data creation, and evaluation difficulty. These conditions matter for generative modelling because they affect what can be trained, how results can be evaluated, and how strongly conclusions can be generalised.

Dedicated African-language resources and models have improved this landscape, but they also show that language coverage, data composition, tokenisation, and evaluation remain central design concerns. Adebara et al. [2023], for example, develop massively multilingual language models for African languages and show that explicit African-language data collection and evaluation can substantially expand coverage beyond the limited African-language support in existing language models. The common implication for this proposal is that African-language modelling requires attention to corpus conditions and evaluation, not only model architecture.

Afrikaans is a meaningful first test case because the NCHLT resource makes a controlled monolingual experiment feasible while still keeping the project within African-language NLP [Eiselen and Puttkammer, 2014]. It must not be treated as representative of all African languages: the results will not automatically transfer to languages with substantially different morphology, orthographic word structure, and tokenisation behaviour [Eiselen and Puttkammer, 2014; Lundin et al., 2025].

Tokenisation is a specific constraint for the proposed research because it determines the sequence length and representation burden faced by the autoencoder and baseline model. Lundin et al. [2025] show, in AfriMMLU evaluations of multilingual LLMs, that higher tokenisation fertility is associated with lower accuracy and greater computational cost. The proposed research will therefore use a SentencePiece BPE tokeniser and explicitly report tokenisation fertility, rather than assuming that the tokeniser is neutral [Kudo and Richardson, 2018; Lundin et al., 2025]. This measurement is needed because a tokeniser that fragments words heavily can increase effective sequence length, make reconstruc-

tion harder, and make baseline comparisons less fair if models do not share the same tokenisation conditions.

Evaluation is also non-trivial. Perplexity, BLEU, chrF, ROUGE, reconstruction loss, and diversity measures each answer different questions. For unconditional generation, there is no single reference output for each generated sample, so reference-overlap metrics such as BLEU, chrF, and ROUGE are treated as secondary or omitted; in the cited diffusion-language work, overlap metrics are primarily used for conditional or reference-based tasks [Gong et al., 2023; Lovelace et al., 2023]. Perplexity under an external reference model can provide one reproducible signal, but it should be interpreted as a limited proxy rather than a semantic quality measure [Lovelace et al., 2023; Shliazhko et al., 2024]. The proposed research therefore combines reconstruction loss, reference-model perplexity, tokenisation fertility, repetition and diversity diagnostics. These choices lead directly into the methodology.

# Chapter 3

# Methodology

The proposed research will evaluate latent diffusion for unconditional Afrikaans text generation using a two-stage methodology. Stage 1 will validate the latent-diffusion language pipeline on English text. Stage 2 will adapt the validated pipeline to the NCHLT Afrikaans Text Corpora and evaluate the Afrikaans results. This sequencing is important because it separates implementation feasibility from Afrikaans-specific corpus, tokenisation, and representation effects.

## 3.1 Hypothesis, Research Questions, Aim, and Objectives

**Hypothesis.** After implementation validation on English text, a latent-diffusion pipeline adapted to unconditional Afrikaans text generation using the NCHLT Afrikaans Text Corpora will produce a latent representation space that is sufficient for reconstruction-quality text generation and will achieve reference-model perplexity close enough to a non-diffusion autoregressive baseline, under matched experimental conditions, to be considered a credible modelling direction for this bounded setting.

In this proposal, **sufficient** means that reconstruction loss and researcher-side inspection of reconstructed samples indicate that the autoencoder preserves enough lexical and syntactic information for diffusion results to be interpretable. **Comparable** means that the Afrikaans latent-diffusion model's mGPT perplexity is no more than 20% higher than the matched autoregressive baseline's mGPT perplexity. This 20% tolerance is a provisional pre-registered decision rule: it distinguishes approximate competitiveness from a clear practical degradation while avoiding the stronger claim that the model has achieved human-level quality. Because there is no published Afrikaans latent-diffusion perplexity threshold, the actual perplexity ratio will be reported alongside this rule so that readers can judge whether the chosen tolerance was too permissive or too strict. The threshold is a rule for interpreting the experiment, not proof of semantic quality. **Credible** means that the model satisfies the reconstruction requirement, is not more than 20% worse than the non-diffusion baseline under the mGPT perplexity rule, and does not show severe degeneracy according to repetition and diversity diagnostics.

The hypothesis is broken into three research questions:

1. **RQ1:** To what extent does the English validation stage identify implementation and scaling constraints that must be controlled before interpreting Afrikaans adaptation results?
2. **RQ2:** To what extent do learned latent representations of Afrikaans text from the NCHLT corpus preserve linguistic information sufficient for reconstruction?
3. **RQ3:** Does the adapted Afrikaans latent-diffusion model compare credibly against a matched autoregressive baseline under mGPT perplexity, diversity/repetition diagnostics, and tokenisation fertility analysis?

**Aim.** The aim of the proposed research is to evaluate whether a latent-diffusion pipeline, first validated on English text, can be adapted into a credible method for unconditional Afrikaans text generation on the selected NCHLT Afrikaans Text Corpora.

**Objectives.** The proposed research has four objectives:

1. To use English validation to identify implementation, scaling, and evaluation constraints that must be fixed before the Afrikaans adaptation is interpreted.
2. To train or adapt the validated pipeline for Afrikaans and evaluate whether its latent representations preserve enough information for reconstruction of held-out NCHLT Afrikaans text.
3. To compare Afrikaans latent-diffusion generations with a matched autoregressive baseline using mGPT perplexity, diversity and repetition diagnostics, and researcher-side sample inspection.
4. To analyse how SentencePiece BPE tokenisation fertility, corpus conditions, and automatic metric limitations affect interpretation of the Afrikaans results.

The English validation stage is a methodological precondition rather than a contribution claim. The hypothesis will be supported if the English stage establishes stable implementation settings, the Afrikaans autoencoder achieves acceptable reconstruction behaviour on held-out text, the Afrikaans latent-diffusion model's mGPT perplexity is no more than 20% higher than the matched autoregressive baseline, and repetition/diversity diagnostics do not indicate obvious degenerate generation. The hypothesis will be weakened or rejected if the implementation constraints cannot be stabilised, if Afrikaans reconstruction is poor, if latent diffusion has more than 20% higher mGPT perplexity than the matched baseline, or if automatic diagnostics show that low perplexity is achieved mainly through repetitive or formulaic output.

## 3.2 Data

The proposed research will use two data stages. The first stage will use OpenWebText as a public English web-text corpus [Gokaslan and Cohen, 2019]. OpenWebText is a web corpus resource rather than a conventional peer-reviewed dataset paper, so the final report will record the live source URL, access date, licence information, and any subset rule used. A documented OpenWebText subset will be used if the full corpus exceeds the available compute budget. The purpose of this corpus is to validate the implementation, training loop, generation procedure, and evaluation scripts, not to make a new contribution about English generation. The final report will record the English corpus source,

licence, download date, number of documents or examples, total words, total tokens, vocabulary size, sequence-length statistics, sampling rule, and split sizes.

The second and main stage will use the NCHLT Afrikaans Text Corpora as the bounded experimental corpus. Eiselen and Puttkammer document NCHLT text resources for South African languages, including Afrikaans corpus and lexical resources [Eiselen and Puttkammer, 2014]. The project will identify the exact text component after corpus inspection; if a clean or normalised corpus component is available, it will be preferred for unconditional monolingual generation because that component is closest to the normalised text required for language modelling.

Exact corpus statistics will not be assumed before inspection. Before training, the project will record the number of documents, total words, total tokens under the SentencePiece BPE tokeniser, vocabulary size, average and median sequence length, licence details, download date, excluded files, cleaning decisions, and final train/validation/test split sizes for both the English validation corpus and the Afrikaans corpus. These statistics will be included in the final report so that the results can be interpreted relative to the actual corpus rather than an assumed data scale.

The proposed research will use a SentencePiece BPE tokeniser because SentencePiece supports language-independent subword training from raw text [Kudo and Richardson, 2018]. This tokeniser choice is separate from the latent-diffusion language architecture adapted from Lovelace et al. [2023]. The English validation stage and the Afrikaans stage will each use a tokeniser trained only on that stage's training split. Within the Afrikaans stage, the same SentencePiece BPE tokeniser will be used for the autoencoder, latent-diffusion pipeline, matched autoregressive baseline, reconstruction analysis, and tokenisation-fertility analysis. This is necessary because tokenisation affects reconstruction difficulty, effective sequence length, tokens-per-word fertility, and fairness of the baseline comparison [Lundin et al., 2025].

The data preparation pipeline will follow these steps:

1. Download OpenWebText and the NCHLT Afrikaans files, recording source URLs, licence information, and download dates.
2. Inspect each corpus structure and identify the text components to be used for training.
3. Remove unusable files, duplicate documents, non-text artifacts, and extremely short fragments according to documented rules.
4. Apply conservative text normalisation consistent with SentencePiece BPE while preserving language-specific orthographic information.
5. Split each corpus into training, validation, and test sets using an 80/10/10 ratio at document level where possible, so that near-identical segments from the same document do not leak across splits.
6. Train the SentencePiece BPE tokeniser using only the relevant training split, then compute corpus statistics for all splits.

Leakage prevention is important because reconstruction and generation metrics can be inflated if validation or test material appears in the training data. The split will there-

fore be created before tokenisation statistics are reported and before any model training begins. If the corpus structure prevents a clean document-level split, the final report will document the fallback splitting rule and the remaining leakage risk.

The minimum viable corpus condition is that preprocessing must yield enough usable, non-duplicate training examples to create train, validation, and test splits and to run at least a reduced autoencoder training pass whose validation loss can be monitored. If preprocessing yields too few usable training examples for stable training, the proposed research will report this as a primary limitation, reduce model capacity where appropriate, and interpret any negative result as evidence about feasibility under the available corpus rather than as a general failure of latent diffusion. The proposed research will not treat the English validation corpus as a contribution or the NCHLT Afrikaans Text Corpora as representative of all Afrikaans text or of African-language text in general. The English corpus is an implementation control; the NCHLT Afrikaans corpus is the bounded corpus for the main study.

## 3.3 Model and Method

The proposed method will use a latent-diffusion pipeline:

$$x \xrightarrow{E} z \xrightarrow{\text{diffusion}} \hat{z} \xrightarrow{D} \hat{x}, \tag{3.1}$$

where $x$ is a text sequence, $E$ is an encoder, $z$ is the latent representation, the diffusion model denoises latent variables, $D$ is a decoder, and $\hat{x}$ is generated text. Figure 3.1 summarises the pipeline.

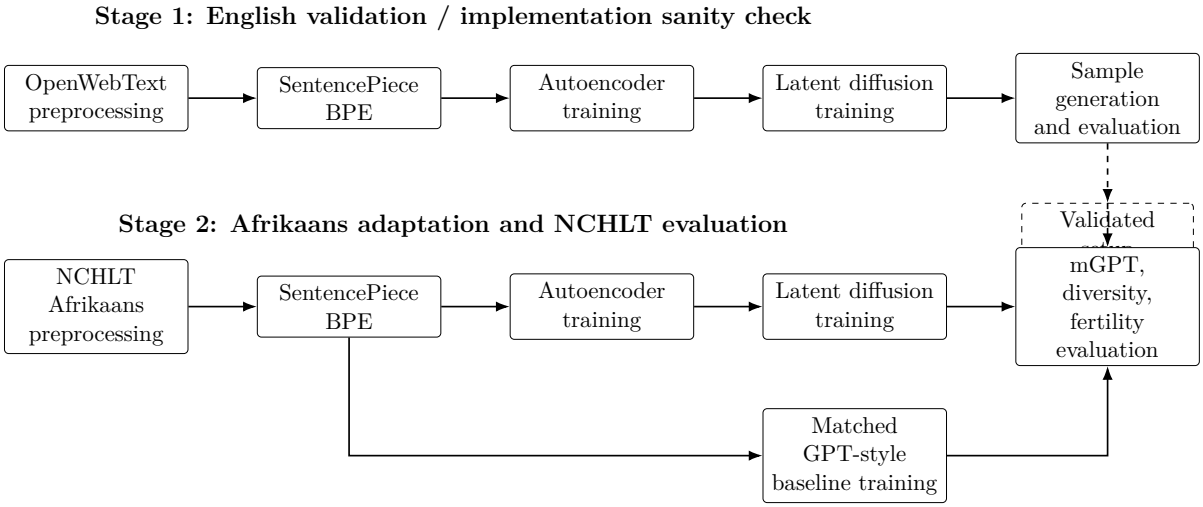



Figure 3.1: Two-stage experimental pipeline for English validation and Afrikaans/NCHLT evaluation. The Afrikaans stage is the focus of the hypothesis and all contribution claims; the English stage provides an implementation control before Afrikaans adaptation.

The primary implementation starting point will be the public repository associated with Lovelace et al. [2023]. That implementation is the closest available match to the proposed pipeline because it includes a language autoencoder, latent diffusion process, and decoding stage. The implementation will first be run on OpenWebText to validate setup and then

adapted to use the NCHLT Afrikaans corpus, the SentencePiece BPE tokeniser, and the compute budget available for the honours project.

The model will have three main components:

1. **Text autoencoder.** The encoder $E$ will map SentencePiece-tokenised text sequences into fixed-size continuous latent representations. The decoder $D$ will reconstruct text from those representations. A 64-dimensional autoencoder latent dimension is used as the initial configuration, matching the monolingual setting reported by Lovelace et al. [2023, Table 13], because it provides a computationally feasible bottleneck for an honours-scale experiment while still allowing the reconstruction objective to test whether the representation retains sufficient sentence-level information. This value is a starting configuration rather than an assumption that Lovelace et al.'s English-scale result transfers unchanged to NCHLT Afrikaans. It will be checked during the English validation stage and adjusted only if reconstruction collapses or compute cost makes the minimum viable experiment infeasible. Following Lovelace et al. [2023], the project uses a non-variational compression-and-reconstruction pipeline rather than a standard sentence VAE; this avoids introducing the VAE KL term associated with posterior-collapse or KL-vanishing pressure [Bowman et al., 2016; Li et al., 2020].
2. **Latent diffusion model.** A denoising model $\epsilon_\theta$ will be trained to reverse noise added to latent representations using the objective in Equation (2.4). The initial diffusion setup will follow Lovelace et al. [2023], including its default cosine noise schedule and the repository's default timestep setting where feasible. If compute is limited, the number of diffusion steps, model width, batch size, or training epochs will be reduced before changing the research question.
3. **Decoder-based generation.** At generation time, latent representations will be sampled from the learned diffusion model and decoded into text using $D$. Generated samples will be saved for repetition/diversity diagnostics.

The minimum viable experiment is an OpenWebText validation run followed by the Afrikaans autoencoder, Afrikaans latent-diffusion model, and matched Afrikaans autoregressive baseline on the same Afrikaans data split and SentencePiece BPE tokeniser. If available compute is lower than expected, optional ablations will be removed before the main Afrikaans comparison is weakened.

## 3.4 Experimental Design

The experiments are designed to separate implementation feasibility, representation quality, diffusion performance, tokenisation behaviour, and baseline comparison. The English stage will be completed first because it checks whether the implementation can run in a relatively well-supported language setting. The Afrikaans stage will then test whether the validated pipeline remains credible under NCHLT Afrikaans corpus conditions.

The experimental procedure will follow these stages:

1. **English implementation validation:** Prepare OpenWebText, run the autoencoder and latent-diffusion training process in full or reduced form, generate sample outputs, and confirm that evaluation scripts execute correctly.

2. **Afrikaans corpus preparation:** Prepare the selected NCHLT Afrikaans corpus component, apply documented preprocessing, and create 80/10/10 train/validation/test splits.

3. **Tokeniser preparation:** Train a SentencePiece BPE tokeniser using the relevant training split only. Record tokens per word, vocabulary size, and sequence-length statistics for English and Afrikaans.

4. **Afrikaans autoencoder training:** Train or adapt the text autoencoder on the Afrikaans training split and evaluate reconstruction loss on validation and test splits.

5. **Afrikaans latent diffusion training:** Train the diffusion model over Afrikaans latent representations extracted from the trained encoder.

6. **Afrikaans baseline training:** Train a lightweight autoregressive Transformer/GPT-style language model on the same Afrikaans split, SentencePiece BPE tokeniser, maximum sequence length, and compute budget.

7. **Evaluation:** Evaluate reconstruction quality, reference-model perplexity, tokenisation fertility, repetition/diversity diagnostics, and generated-sample errors using the metrics in Section 3.6.

The main controlled comparison is between the Afrikaans latent-diffusion model and the Afrikaans autoregressive baseline. The English validation stage is not used to claim that English and Afrikaans performance are directly comparable. Instead, it provides a control against implementation failure. If the English stage works but the Afrikaans stage fails, the analysis can focus more credibly on representation quality, tokenisation, corpus conditions, or evaluation limitations.

## 3.5 Baselines and Controls

Baselines and controls are required because latent diffusion cannot be evaluated in isolation. The proposed study will use the following comparison points:

1. **English implementation control.** The public latent-diffusion language pipeline will first be run on English text to verify that training, generation, and evaluation execute correctly before Afrikaans adaptation.

2. **Afrikaans autoencoder-only control.** Held-out Afrikaans text will be encoded and decoded without diffusion. This tests whether the latent representation is useful before diffusion is added.

3. **Matched Afrikaans autoregressive baseline.** A lightweight GPT-style autoregressive Transformer will be trained on the same Afrikaans training split, SentencePiece BPE tokeniser, maximum sequence length, and compute budget as the proposed latent-diffusion system. This is the primary baseline for generation quality.

4. **Reference text score.** mGPT [Shliazhko et al., 2024] will score held-out real NCHLT Afrikaans test text to provide a limited external reference point for the perplexity scale, following the use of external language-model perplexity as one au-

tomatic generation diagnostic [Lovelace et al., 2023]. This is not a trained baseline; it shows how the external scoring model treats real text from the same corpus.

Table 3.1: Baseline and control roles in the proposed research.

| Role | Option | Purpose |
|---|---|---|
| Implementation control | OpenWebText latent-diffusion validation run | Confirms that the pipeline can train, generate, and be evaluated before Afrikaans adaptation. |
| Main method | Latent diffusion for language adapted from Lovelace et al. [2023] | Tests whether latent diffusion is credible for Afrikaans/NCHLT generation. |
| Autoencoder-only control | Afrikaans encoder-decoder without diffusion | Tests whether representation quality is sufficient before diffusion. |
| Primary baseline | Matched Afrikaans GPT-style autoregressive Transformer | Tests whether latent diffusion is competitive against a simpler non-diffusion generator under matched data and tokenisation conditions. |
| Reference text score | mGPT scoring real held-out NCHLT text | Provides a limited external perplexity reference point, not a target expected to be matched. |

## 3.6 Metrics and Evaluation

All evaluation will use existing corpora, model outputs, automatic metrics, and researcher-side inspection of generated samples.

**English validation evidence** will be used for RQ1. The English stage will be considered fully successful if the public pipeline can train or run in a reduced configuration on OpenWebText, generate samples, and produce evaluation outputs without unresolved implementation failures. It will be considered a partial success if preprocessing, tokenisation, generation, and evaluation work but training must be substantially reduced for compute reasons. The English validation record will document which settings had to be fixed or reduced, including tokeniser settings, maximum sequence length, latent dimension, batch size, diffusion steps, and evaluation-script assumptions. The English results will be reported as validation evidence, not as a claim that English performance predicts Afrikaans performance.

**Reconstruction loss** is the primary metric for Afrikaans representation quality, following the autoencoding role of the latent language-generation pipeline in Lovelace et al. [2023]. It measures the loss incurred when text is encoded and decoded:

$$\text{original text} \rightarrow z = E(\text{text}) \rightarrow \hat{\text{text}} = D(z).$$

Reconstruction loss will be reported for validation and test sets and used as a gate before strong claims are made about generation quality. The autoencoder-only control will be

treated as adequate only if validation reconstruction loss improves during training and researcher-side inspection shows that decoded samples preserve basic lexical and syntactic content from the inputs. Researcher-side inspection of reconstructions will use three fixed checks: whether major named entities or content words are preserved, whether the main clause structure remains recognisable, and whether the reconstruction introduces obvious hallucinated lexical material not supported by the input. If reconstruction loss remains high or reconstructed samples are visibly unfaithful to the input under these checks, the latent representation will be treated as insufficient for interpreting later diffusion results. This criterion is provisional because no published Afrikaans latent-diffusion reconstruction threshold is available.

**Reference-model perplexity** is the primary metric for Afrikaans generation quality in RQ3. Following the use of external language-model perplexity as one automatic diagnostic for generated text [Lovelace et al., 2023], a fixed pretrained multilingual model, mGPT [Shliazhko et al., 2024], will compute perplexity for generated sequences from the Afrikaans latent-diffusion model, generated sequences from the Afrikaans autoregressive baseline, autoencoder reconstructions, and real held-out NCHLT test text. mGPT is used because it provides a fixed external multilingual language-model score and avoids evaluating the latent-diffusion model using its own training objective. Before the generated-text comparison is interpreted, the proposed research will check whether mGPT assigns stable and interpretable perplexity to real held-out NCHLT Afrikaans text across sentence-length bands. If mGPT perplexity on real Afrikaans text is anomalously high or unstable, the perplexity comparison will be reported as a limitation and interpreted only as a weak diagnostic rather than as the primary generation-quality result. The interpretation rule is that latent diffusion is not more than 20% worse than the matched baseline if its mGPT perplexity is no more than 20% higher than the baseline's mGPT perplexity. This 20% rule is an experimental decision rule, not proof of human-level quality. The actual percentage gap will be reported even when the decision rule is met, preventing the threshold from hiding a practically important difference. mGPT perplexity is also not a perfect measure of semantic quality for Afrikaans, and a model can receive a good perplexity score by producing repetitive or generic output [Lovelace et al., 2023; Shliazhko et al., 2024].

**Repetition and diversity diagnostics** will be used to interpret perplexity. The proposed research will compute distinct-1 and distinct-2, which measure the proportion of unique unigrams and bigrams in generated text, and repeated n-gram rates, which measure whether generated outputs collapse into repeated phrases [Gong et al., 2023; Lovelace et al., 2023]. These diagnostics do not replace perplexity, but they test whether a low perplexity score is accompanied by degeneracy. A model with acceptable perplexity but very low diversity or high repeated n-gram rates will not be considered credible.

**Tokenisation fertility** will be measured as tokens per word under the SentencePiece BPE tokeniser [Kudo and Richardson, 2018; Lundin et al., 2025]. The analysis will report average and distributional fertility values for English and Afrikaans, then focus on whether observed Afrikaans fertility co-occurs with higher reconstruction loss, higher reference-model perplexity, or longer effective sequence lengths. This metric answers RQ3 by connecting tokenisation behaviour to representation quality, baseline comparability, and compute cost.

**Researcher-side sample inspection** will supplement the automatic metrics. Generated samples from each model condition will be inspected with a fixed error checklist covering unreadable output, repeated fragments, broken Afrikaans morphology, excessive copying from the training corpus, abrupt truncation, and obvious domain-inappropriate text. The inspection notes will be kept with the generated sample files so that the automatic scores can be interpreted against visible failure modes.

Accuracy is not appropriate because the task is not classification. The metrics together provide diagnostic evidence about implementation, representation, and degeneration, but they do not prove semantic quality.

## 3.7 Research Question and Metric Alignment

Table 3.2 summarises how the hypothesis, research questions, experiments, evidence, and decision rules connect.

Table 3.2: Alignment between the research questions, experiments, evidence, and interpretation rules.

| **Research question** | **Experiment, data, and control** | **Metric or evidence** | **Decision rule** |
|---|---|---|---|
| RQ1: What implementation and scaling constraints must be controlled before Afrikaans adaptation is interpreted? | Run the public latent-diffusion language pipeline on OpenWebText as an implementation control before Afrikaans adaptation. | Full or partial run status; generated samples; evaluation outputs; documented setup issues; fixed or reduced settings carried into the Afrikaans stage. | The English stage is adequate only if it identifies stable preprocessing, tokenisation, model, and evaluation settings for the Afrikaans stage. Failure occurs if the implementation cannot be made operational or if unresolved setup issues make later Afrikaans failures uninterpretable. |
| RQ2: Does the Afrikaans latent space preserve information? | Train the Afrikaans autoencoder on selected NCHLT train/validation/test splits and reconstruct held-out text using the autoencoder-only control. | Validation and test reconstruction loss; researcher-side inspection of reconstructed samples using fixed lexical, syntactic, and hallucination checks. | The representation is adequate only if validation loss improves during training and decoded samples preserve basic lexical and syntactic content under the fixed inspection checks. High loss, unreadable reconstructions, or evidence that the latent representation is not used weakens the hypothesis. |
| RQ3: Is generated text comparable to a non-diffusion baseline, and how does fertility affect results? | Generate samples from Afrikaans latent diffusion and a matched GPT-style baseline using the same NCHLT splits and SentencePiece BPE tokeniser. | mGPT perplexity after real-text stability checking; Distinct-1/Distinct-2; repeated n-gram rate; tokens per word; mGPT score on real held-out NCHLT text. | The latent-diffusion model supports the hypothesis if mGPT is stable enough to interpret, its perplexity is no more than 20% higher than the matched baseline, and it shows no severe repetition collapse. More than 20% higher mGPT perplexity, unstable mGPT scores on real Afrikaans text, repetitive output, or confounded fertility analysis weakens the hypothesis. |

# Chapter 4

# Research Plan

This chapter outlines the proposed timeline, deliverables, risks, mitigation strategies, and limitations. The plan assumes a 17-week active research period after proposal approval and prioritises the minimum viable experiment: English implementation validation, Afrikaans data preparation, autoencoder evaluation, latent-diffusion training, matched autoregressive baseline training, automatic evaluation, and final analysis. Writing will be done iteratively throughout the project.

## 4.1 Timeline

Table 4.1 presents the proposed timeline. The plan is feasible if GPU access is available. If compute is limited, model width, latent dimension, sequence length, batch size, diffusion steps, and optional ablations will be reduced while preserving the OpenWebText validation stage and the core Afrikaans latent-diffusion/baseline comparison. The riskiest period is weeks 6–11, so the diffusion-training block includes an internal buffer: if autoencoder training or corpus preparation overruns, the first part of week 9 will be used to stabilise that stage before the diffusion run is scaled down.

Table 4.1: Proposed research timeline.

| Period | Planned work |
|---|---|
| Weeks 1–2 | Set up the public latent-diffusion language implementation. Document OpenWebText and NCHLT access. Run a reduced OpenWebText validation pass and verify evaluation scripts. Start the methods log and report skeleton. |
| Weeks 3–5 | Build preprocessing pipeline, create document-level 80/10/10 splits where possible, train SentencePiece BPE tokenisers, and record fertility and sequence-length statistics for English and Afrikaans. |
| Weeks 6–8 | Adapt the Afrikaans text autoencoder. Train the initial 64-dimensional configuration, evaluate validation reconstruction loss, inspect reconstructions, and decide whether the latent space is adequate for diffusion training. |
| Weeks 9–11 | Train the Afrikaans latent-diffusion model over learned latent representations, using the first part of week 9 as buffer if autoencoder training or corpus preparation overruns. Reduce model scale if compute is insufficient while preserving the main research questions. Maintain experiment logs and preliminary result tables. |
| Weeks 12–13 | Train the matched Afrikaans autoregressive baseline and run the autoencoder-only control. Evaluate reconstruction loss, mGPT perplexity, Distinct-1/Distinct-2, repeated n-gram rate, and tokenisation fertility. |
| Weeks 14–15 | Hold a supervisor-feedback checkpoint. Inspect generated samples, analyse failure modes, revise interpretation, and complete diagnostic tables. |
| Weeks 16–17 | Complete final interpretation, reproducibility checks, code/configuration cleanup, report revision, bibliography audit, and final LaTeX compilation. |

## 4.2 Planned Deliverables

The planned deliverables are concrete artefacts, not the same as the proposed contributions:

1. An OpenWebText validation record showing corpus selection, setup steps, generated samples, and evaluation-script outputs.
2. Preprocessing scripts for OpenWebText and the selected NCHLT Afrikaans corpus component, including cleaning scripts, split files, and corpus-statistics output.
3. SentencePiece BPE tokeniser artefacts and tokenisation statistics for English and Afrikaans, including tokens per word, vocabulary size, and sequence-length distribution.
4. A trained Afrikaans text autoencoder with validation and test reconstruction-loss results.
5. A trained Afrikaans latent-diffusion model for unconditional Afrikaans text generation.
6. A matched Afrikaans lightweight autoregressive baseline, baseline configuration files, and an Afrikaans autoencoder-only control.
7. Evaluation scripts for reconstruction loss, mGPT perplexity, distinct-1, distinct-2, repeated n-gram rate, and tokenisation fertility.

8. Generated sample files for each model condition and researcher-side error-analysis notes.
9. Results tables and interpretation notes mapping each metric back to the relevant research question.
10. A final written report with reproducible code and configuration files.

## 4.3 Risk Factors and Mitigation Strategies

The proposed research has five main risks.

**Risk 1: The English validation stage may not run cleanly.** If the public implementation cannot be made operational on OpenWebText, the Afrikaans stage would be delayed and failures could be implementation-related rather than research-related. The mitigation is to start with a reduced English run, document setup issues early, reduce model size if necessary, and treat a working reduced pipeline as sufficient for moving to Afrikaans.

**Risk 2: Weak or collapsed Afrikaans latent representation.** If the Afrikaans autoencoder fails to preserve useful information, diffusion in latent space will not be interpretable. This is motivated by posterior-collapse and KL-vanishing concerns in text autoencoders [Bowman et al., 2016; Li et al., 2020]. The mitigation is to prefer a non-variational compression-and-reconstruction pipeline following Lovelace et al. [2023], monitor reconstruction loss early, inspect reconstructed samples, reduce decoder strength if needed, and treat poor reconstruction as an empirical result rather than hiding it.

**Risk 3: Corpus size, domain, or data quality may limit Afrikaans training.** The selected NCHLT Afrikaans corpus component may be too small, repetitive, domain-specific, or noisy for reliable training from scratch. The mitigation is to document corpus statistics before training, remove duplicates, use conservative preprocessing, and reduce model size. If preprocessing yields too few usable training examples for stable training, the proposed research will report this as a primary feasibility limitation rather than treating the result as a general failure of latent diffusion.

**Risk 4: Compute limitations may prevent full-scale training.** Training an autoencoder, diffusion model, and baseline can exceed available GPU resources. The mitigation is to define the minimum viable experiment in advance, reduce model width, latent dimension, batch size, maximum sequence length, and diffusion steps before dropping the core Afrikaans comparison, and remove optional ablations before weakening the hypothesis test.

**Risk 5: Automatic metrics may constrain interpretation.** mGPT perplexity can reward repetitive or generic text, and automatic metrics do not prove semantic quality [Lovelace et al., 2023; Shliazhko et al., 2024]. The mitigation is to report perplexity together with Distinct-1/Distinct-2, repeated n-gram rates, tokenisation fertility, and documented researcher-side sample inspection, while explicitly limiting the conclusions supported by these metrics.

## 4.4 Limitations

The proposed research has clear boundaries.

First, the English validation stage is not a claim about English text generation and is not the main contribution. It is a methodological control used to check that the implementation works before Afrikaans adaptation. A successful English run does not imply that the Afrikaans model will work, because the data conditions, tokeniser behaviour, and language-specific representation issues are different.

Second, the main study focuses on Afrikaans and the NCHLT Afrikaans Text Corpora. Positive results would not justify claims about all African languages. Afrikaans differs from many African languages in resource availability, morphology, orthography, and sociolinguistic context [Nekoto et al., 2020; Eiselen and Puttkammer, 2014; Lundin et al., 2025]. The study can provide evidence about one bounded Afrikaans corpus setting, not a general theory of African-language generation.

Third, the results will not generalise to all Afrikaans domains. The NCHLT corpus has its own composition and collection history. A model that performs acceptably on this corpus may not perform acceptably on social media, conversational Afrikaans, educational text, legal text, or other specialised domains.

Fourth, the task is unconditional text generation. Results from this proposal will not automatically transfer to translation, summarisation, simplification, dialogue, question answering, or deployable language systems. Those tasks require different datasets, controls, and metrics.

Fifth, automatic metrics cannot fully prove semantic quality, factuality, stylistic suitability, or social usefulness. Reference-model perplexity, reconstruction loss, fertility, and diversity diagnostics can provide reproducible evidence, but they cannot establish every aspect of language quality [Lovelace et al., 2023; Shliazhko et al., 2024]. The proposed research remains useful despite this limitation because it tests whether the latent-diffusion pipeline is viable enough to merit further, broader evaluation.

Sixth, no human evaluation is performed. Future work with appropriate ethics clearance could include participant-based fluency, adequacy, or preference evaluation, but that is outside the scope of this proposed project.

# Chapter 5

# Conclusion

This proposal sets out a plan to evaluate latent diffusion for unconditional Afrikaans text generation using the selected NCHLT Afrikaans Text Corpora. The central motivation is that diffusion-based language generation depends on representation choices [Li et al., 2022; Lovelace et al., 2023], and the cited African-language NLP literature shows that corpus, tokenisation, and evaluation constraints shape what can be learned and evaluated [Joshi et al., 2020; Nekoto et al., 2020; Lundin et al., 2025]. The proposal therefore treats latent representation quality as the main point of investigation, not as an implementation detail that can be assumed to work.

The proposed research will proceed in two stages. It will first validate the latent-diffusion language pipeline on OpenWebText, because the public implementation and much of the existing literature are grounded in English or high-resource settings. This stage is necessary to reduce implementation uncertainty before the pipeline is adapted to Afrikaans. The project will then test the main hypothesis: that the validated pipeline can be adapted to NCHLT Afrikaans, learn a representation space sufficient for reconstruction-quality text generation, and produce generated text whose mGPT perplexity is no more than 20% higher than a matched autoregressive baseline while avoiding severe repetition collapse.

This hypothesis matters because it is falsifiable and diagnostic. A positive result would show that, for this bounded Afrikaans/NCHLT setting, a latent representation can be learned well enough for latent diffusion to produce text that is approximately competitive with a simpler non-diffusion baseline under automatic diagnostics. A negative result would still contribute evidence about where the pipeline breaks. For example, if the autoencoder fails while the matched autoregressive baseline is reasonable, the evidence would suggest that the bottleneck is the latent representation stage rather than simply the diffusion sampler. That distinction matters for future African-language diffusion work, because it would indicate whether researchers should prioritise representation learning before scaling diffusion models.

Unconditional generation is a defensible starting task because it avoids requiring parallel or reference outputs and focuses the project on whether the latent space, diffusion model, tokeniser, and automatic diagnostics function coherently in one controlled monolingual setting.

The proposed contributions are bounded. First, the project will provide an empirical evaluation of latent diffusion for unconditional Afrikaans generation using one documented corpus, after English implementation validation. Second, it will analyse how representation quality, tokenisation fertility, data conditions, and automatic metric limitations affect the interpretation of latent-diffusion results. Third, it will analyse how these factors interact within a reproducible diagnostic framework containing preprocessing scripts, tokeniser artefacts, model configurations, baseline settings, evaluation scripts, generated samples, results tables, and analysis notes. These contributions are useful even if the main hypothesis is weakened, because a well-explained negative result can prevent later work from treating latent diffusion as a simple plug-in method for low-resource language generation.

Machine-learning and NLP researchers should care about this work because it tests a representation-centred generative method outside the high-resource settings where many modelling assumptions are developed. The cited African-language NLP and tokenisation literature shows that resource availability, tokenisation, benchmark coverage, and compute costs are not neutral background conditions; they shape which methods can be built and evaluated [Joshi et al., 2020; Nekoto et al., 2020; Lundin et al., 2025]. A careful OpenWebText-to-Afrikaans experimental sequence can therefore provide evidence about what changes when a latent-diffusion language pipeline moves from a better-supported language setting to a bounded African-language corpus setting.

Broader African-language NLP work should care because language technology for African languages remains unevenly resourced and unevenly evaluated [Joshi et al., 2020; Nekoto et al., 2020; Adebara et al., 2023]. Afrikaans is only one case study and is relatively better resourced than many African languages [Nekoto et al., 2020; Eiselen and Puttkammer, 2014], so the results cannot be generalised to all African languages or to languages with substantially different morphology and orthographic word structure, such as isiZulu or isiXhosa [Eiselen and Puttkammer, 2014; Lundin et al., 2025]. The value of the proposed research is that it examines one concrete setting carefully, avoids overclaiming, and reports what can and cannot be concluded from automatic evaluation. It contributes diagnostic evidence and reproducible artefacts, not a universal solution. Follow-on work could test a related NCHLT setting for isiZulu or isiXhosa, add participant-based fluency or preference evaluation after ethics clearance, or move from unconditional generation to a conditional task such as translation or summarisation with task-specific baselines. The final project will therefore evaluate whether latent diffusion is a credible modelling direction for this specific Afrikaans corpus and task, while making clear where further work on other languages, domains, and tasks would be required.

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